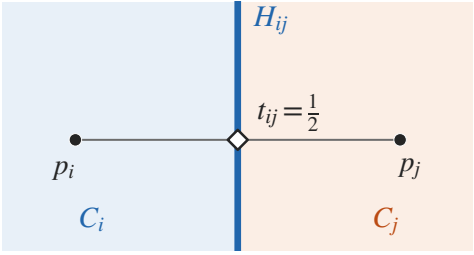
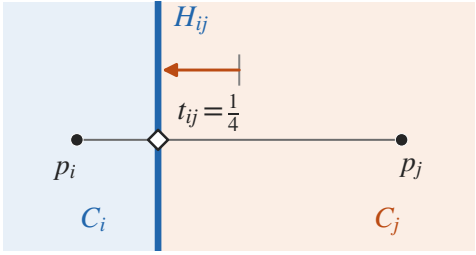
(a) Equal weights



$(w_i, w_j) = (0, 0)$

(b) Increasing w_j



$(w_i, w_j) = (0, 2)$